

PROJECT REPORT: SMART STUDENT MANAGEMENT SYSTEM

Department of Computer Science & Engineering | Academic Year 2024-25

Submitted by: Rahul Sharma (Roll: CS2024-042)

Faculty Guide: Dr. Priya Nair

1. ABSTRACT

The Smart Student Management System (SMS) is an automated web-based platform designed to streamline ac

2. PROBLEM DEFINITION & MOTIVATION

In modern higher education institutions, administrative staff and faculty spend up to 14 hours weekly managing c

- Lack of centralized real-time synchronization between department portals.
- Vulnerabilities in paper-based viva evaluations and mark tabulation.
- Inability to quickly generate compliance reports for academic accreditation committees.

The proposed system automates end-to-end student lifecycle management with high data integrity.

3. LITERATURE REVIEW & RELATED WORK

We surveyed several existing Enterprise Resource Planning (ERP) frameworks:

- Sharma et al. (2021) designed a cloud-hosted university information system but noted significant database latency.
- Patel & Verma (2023) implemented attendance tracking using RFID tokens; however, hardware deployment cost was high.
- Kumar (2022) surveyed web-based grading portals, identifying a critical gap in automated rubric-based faculty evaluation.

4. METHODOLOGY & PROPOSED SYSTEM ARCHITECTURE

The system is architected into three decoupled layers:

- Presentation Layer: Responsive Single Page Application (SPA) built in React with Tailwind CSS.
- Application Layer: RESTful microservices powered by Node.js/Express, managing authentication via JWT.
- Data Storage Layer: Normalized Relational Database (3NF) ensuring ACID transactional compliance for grade management.
- Security Subsystem: Bcrypt hashing for password credentials and token-based RBAC for Student, Faculty, and Admin roles.

5. IMPLEMENTATION & TECHNICAL STACK

- Frontend Modules: Student Portal, Faculty Grading Workbench, Admin User Roster.
- Backend Services: Student Enrollment API, Marks Processing Controller, Report Generation Service.
- Database Tables: Students, Courses, Enrollments, Grades, RubricCriteria, AuditLogs.

6. RESULTS & PERFORMANCE BENCHMARKS

The prototype was tested across a simulated dataset of 500 student records:

- User Authentication Latency: Average response time of 42ms under 50 concurrent requests.
- Transcript Generation: PDF generation latency averaged 180ms per student record.
- Security Validation: SQL injection tests and XSS payload validations passed without exploitability.

7. CONCLUSION & FUTURE SCOPE

The Smart Student Management System demonstrates an efficient, user-friendly, and secure platform for academic institutions.

- Integrating biometrics or AI facial recognition for attendance verification.
- Deploying machine learning models for early predictive warnings on at-risk student performance.

8. REFERENCES

- [1] Sharma, R., et al. (2021). "Cloud-based Campus Management Systems." IEEE Trans. Edu.
- [2] Patel, K., & Verma, S. (2023). "Automated Attendance Systems." Int. J. Computer Applications.
- [3] Kumar, A. (2022). "Standardized Grading Portals in Higher Education." ACM SIGCSE.